

# SIMATS ENGINEERING

## ASSESSMENT TOOL 2

**COURSE NAME:** CSA16 **COURSE CODE:** Data Warehousing and Data Mining

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### COURSE OUTCOME COVERED

**CO4:** Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 40%

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**QUESTIONS: 05**

**Total Marks:50**

Annexure A

Industry Problem Solving Task

| <i>Criteria</i>  | Understanding of Industry Problem (2) | Application of Engineering Concepts (2.5) | Solution Design (2.5) | Use of Tools (2) | Analysis (1) | <i>Total(10)</i> |
|------------------|---------------------------------------|-------------------------------------------|-----------------------|------------------|--------------|------------------|
| <i>Weightage</i> | 20%                                   | 25%                                       | 25%                   | 20%              | 10%          | <i>100%</i>      |
| <i>Q1</i>        |                                       |                                           |                       |                  |              |                  |
| <i>Q2</i>        |                                       |                                           |                       |                  |              |                  |
| <i>Q3</i>        |                                       |                                           |                       |                  |              |                  |
| <i>Q4</i>        |                                       |                                           |                       |                  |              |                  |
| <i>Q5</i>        |                                       |                                           |                       |                  |              |                  |

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***Code of Conduct:***

*I \_\_\_\_\_ E.Lohitha(192525056) \_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student: lohitha(192525056)***

## RUBRICS FOR CALCULATION:

| Criteria                            | Weightage | Level 4 – Exemplary                                                          | Level 3 – Proficient                                 | Level 2 – Developing                           | Level 1 – Beginning                               |
|-------------------------------------|-----------|------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------|---------------------------------------------------|
| Criteria                            | Weightage | Level 4 – Exemplary                                                          | Level 3 – Proficient                                 | Level 2 – Developing                           | Level 1 – Beginning                               |
| Understanding of Industry Problem   | 20%       | Comprehensive understanding of real-world problem and constraints            | Good understanding with minor gaps in requirements   | Basic understanding; some requirements unclear | Poor understanding; misinterprets problem context |
| Application of Engineering Concepts | 25%       | Applies advanced and relevant concepts effectively to solve the problem      | Applies appropriate concepts with minor errors       | Limited application of concepts                | Incorrect or no application of concepts           |
| Solution Design                     | 25%       | Innovative, practical, and feasible solution aligned with industry standards | Logical and feasible solution with minor gaps        | Basic solution with limited feasibility        | Unclear or impractical solution                   |
| Use of Tools                        | 15%       | Effective use of modern tools, technologies, and real data                   | Appropriate use of tools with correct implementation | Limited or inconsistent use of tools           | Minimal or incorrect use of tools                 |
| Analysis                            | 10%       | Thorough analysis with validated results and performance evaluation          | Good analysis with reasonable validation             | Basic analysis with limited validation         | Poor or no analysis                               |
| Professionalism                     | 5%        | Highly professional, well-documented, and clearly presented work             | Good documentation and presentation                  | Basic documentation with some gaps             | Poor documentation and presentation               |

## Annexure A

Q1 . Dataset: Supermarket Purchases

**Customer Visits/Month Avg Spend (₹) Items Bought**

|    |    |      |    |
|----|----|------|----|
| C1 | 10 | 2000 | 15 |
| C2 | 4  | 5000 | 8  |
| C3 | 12 | 1800 | 18 |
| C4 | 3  | 6000 | 6  |
| C5 | 11 | 2100 | 16 |

**Question:**

Using the above dataset, propose a clustering approach to segment customers and explain how it supports targeted marketing. (BL3 – 7 Marks)

The screenshot shows the Weka Explorer interface with the SimpleKMeans clustering algorithm applied to the Supermarket Purchases dataset. The 'Clusterer' tab is selected, and the 'SimpleKMeans' algorithm is chosen. The 'Cluster mode' is set to 'Use training set'. The 'Clusterer output' pane displays the following information:

- Number of iterations: 3
- Within cluster sum of squared errors: 3.1081506676744777
- Initial starting points (random):
- Cluster 0: C4,3,6000,6
- Cluster 1: C2,4,5000,8
- Missing values globally replaced with mean/mode
- Final cluster centroids:

| Attribute     | Full Data | Cluster# | 0         | 1 |
|---------------|-----------|----------|-----------|---|
|               | (5.0)     | (2.0)    | (3.0)     |   |
| Customer      | C1        | C2       | C1        |   |
| Visits/Month  | 8         | 3.5      | 11        |   |
| Avg Spend (?) | 3380      | 5500     | 1966.6667 |   |
| Items Bought  | 12.6      | 7        | 16.3333   |   |

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

|   |          |
|---|----------|
| 0 | 2 ( 40%) |
| 1 | 3 ( 60%) |

The status bar at the bottom shows 'Status OK' and a 'Log' button.

Q2. Dataset: Credit Card Transactions

**Transaction Amount (₹) Location Time (hrs)**

|    |      |         |    |
|----|------|---------|----|
| T1 | 200  | Chennai | 10 |
| T2 | 5000 | Delhi   | 2  |
| T3 | 150  | Chennai | 11 |
| T4 | 7000 | Mumbai  | 1  |

### Transaction Amount (₹) Location Time (hrs)

T5                      180                      Chennai 12

#### Question:

Design a clustering-based solution using this dataset to detect unusual or fraudulent transactions. (BL4 – 7 Marks)

The screenshot shows the Weka Explorer interface with the SimpleKMeans clustering algorithm applied. The 'Clusterer' tab is active, and the 'SimpleKMeans' algorithm is selected. The 'Cluster mode' is set to 'Use training set'. The 'Clusterer output' pane displays the following information:

```
Number of iterations: 3
Within cluster sum of squared errors: 4.063311626771005

Initial starting points (random):
Cluster 0: T4,7000,Mumbai,1
Cluster 1: T2,5000,Delhi,2

Missing values globally replaced with mean/mode

Final cluster centroids:
Attribute      Full Data      Cluster#
              (5.0)         (2.0)         (3.0)
=====
Transaction    T1             T2             T1
Amount (?)      2506           6000          176.6667
Location       Chennai        Delhi          Chennai
Time (hrs)      7.2            1.5            11

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances
0      2 ( 40%)
1      3 ( 60%)
```

### Q3. Dataset: Telecom Usage

#### User Call Duration (min) Data Usage (GB) Recharge (₹)

|    |     |     |     |
|----|-----|-----|-----|
| U1 | 300 | 5   | 200 |
| U2 | 100 | 2   | 100 |
| U3 | 350 | 6   | 220 |
| U4 | 80  | 1   | 90  |
| U5 | 320 | 5.5 | 210 |

#### Question:

Explain how clustering can be applied to identify churn-prone users and improve customer

retention strategies. (BL4 – 7 Marks)

The screenshot shows the Weka Explorer interface with the SimpleKMeans clustering algorithm applied to a dataset. The 'Cluster mode' is set to 'Use training set'. The 'Clusterer output' pane displays the following information:

- Number of iterations: 3
- Within cluster sum of squared errors: 3.074911783724699
- Initial starting points (random):
- Cluster 0: U4,80,1,90
- Cluster 1: U2,100,2,100
- Missing values globally replaced with mean/mode
- Final cluster centroids:

| Attribute           | Full Data | Cluster# | 0        | 1     |
|---------------------|-----------|----------|----------|-------|
|                     | (5.0)     |          | (2.0)    | (3.0) |
| User                | U1        | U2       | U1       |       |
| Call Duration (min) | 230       | 90       | 323.3333 |       |
| Data Usage (GB)     | 3.9       | 1.5      | 5.5      |       |
| Recharge (?)        | 164       | 95       | 210      |       |

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

|   |          |
|---|----------|
| 0 | 2 ( 40%) |
| 1 | 3 ( 60%) |

The 'Result list' on the left shows three entries for SimpleKMeans, with the most recent one (14:31:11) selected.

Q4 . Dataset: E-commerce Behavior

**User Product Views Clicks Purchases**

|    |    |    |   |
|----|----|----|---|
| U1 | 50 | 20 | 5 |
| U2 | 30 | 10 | 2 |
| U3 | 60 | 25 | 6 |
| U4 | 20 | 8  | 1 |
| U5 | 55 | 22 | 5 |

**Question:**

Suggest a clustering technique to group users and explain how it improves recommendation

systems. (BL3 – 7 Marks)

The screenshot shows the Weka Explorer interface with the 'Cluster' tab selected. The 'SimpleKMeans' algorithm is chosen with the following parameters: -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance" -R first-last" -I 500 -num-slots 1. The 'Cluster mode' is set to 'Use training set'. The 'Ignore attributes' list is empty. The 'Start' button is visible. The 'Clusterer output' pane shows the following results:

```

Number of iterations: 2
Within cluster sum of squared errors: 2.9243771626297583

Initial starting points (random):

Cluster 0: U3,60,25,6
Cluster 1: U1,50,20,5

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute      Full Data      Cluster#
              (4.0)        (1.0)        (3.0)
=====
User            U1            U3            U1
Product Views   40            60           33.3333
Clicks          15.75         25           12.6667
Purchases       3.5           6            2.6667

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0      1 ( 25%)
1      3 ( 75%)
  
```

The 'Result list' on the left shows four entries for 'SimpleKMeans' at different times, with the most recent one (14:38:09) selected.

Q5. Dataset: Traffic Data

**Location Vehicle Count Avg Speed (km/h) Delay (min)**

|    |     |    |    |
|----|-----|----|----|
| L1 | 200 | 30 | 10 |
| L2 | 500 | 15 | 25 |
| L3 | 220 | 28 | 12 |
| L4 | 600 | 10 | 30 |
| L5 | 210 | 32 | 9  |

**Question:**

Design a clustering-based solution to identify traffic congestion zones and explain how it

helps in traffic management. (BL4 – 7 Marks)

Weka Explorer

Preprocess   Classify   **Cluster**   Associate   Select attributes   Visualize

Clusterer: Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance" -R first-last" -I 500 -num-slots 1

Cluster mode

☒ Use training set  
☐ Supplied test set Set...  
☐ Percentage split % 66  
☐ Classes to clusters evaluation (Num) Delay (min) ∨  
☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

- 14:13:46 - SimpleKMeans
- 14:25:00 - SimpleKMeans
- 14:31:11 - SimpleKMeans
- 14:38:09 - SimpleKMeans
- 14:40:08 - SimpleKMeans**

Clusterer output

Number of iterations: 3  
Within cluster sum of squared errors: 3.113782053684652

Initial starting points (random):  
Cluster 0: L4,600,10,30  
Cluster 1: L2,500,15,25

Missing values globally replaced with mean/mode

Final cluster centroids:

| Attribute        | Full Data | Cluster# |         |
|------------------|-----------|----------|---------|
|                  |           | 0        | 1       |
|                  | (5.0)     | (2.0)    | (3.0)   |
| =====            |           |          |         |
| Location         | L1        | L2       | L1      |
| Vehicle Count    | 346       | 550      | 210     |
| Avg Speed (km/h) | 23        | 12.5     | 30      |
| Delay (min)      | 17.2      | 27.5     | 10.3333 |

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

|   |          |
|---|----------|
| 0 | 2 ( 40%) |
| 1 | 3 ( 60%) |

Status OK Log x 0